ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)	**Residue Upgradation Facility (RUF)** **EPCC-3 Package for** **Visakh Refinery Modernization Project (VRMP)**	L&T Hydrocarbon Engineering **L&T- CHIYODA LIMITED**
Title: (504-LZ-930A/B) CIVIL DRAWING FOR DOCK LEVELER **Doc. No.: BO16-RUF-LT-504-MP-DW-NTPL(1)-13005**		**Rev. No.:** **Page No: 1 of 4**

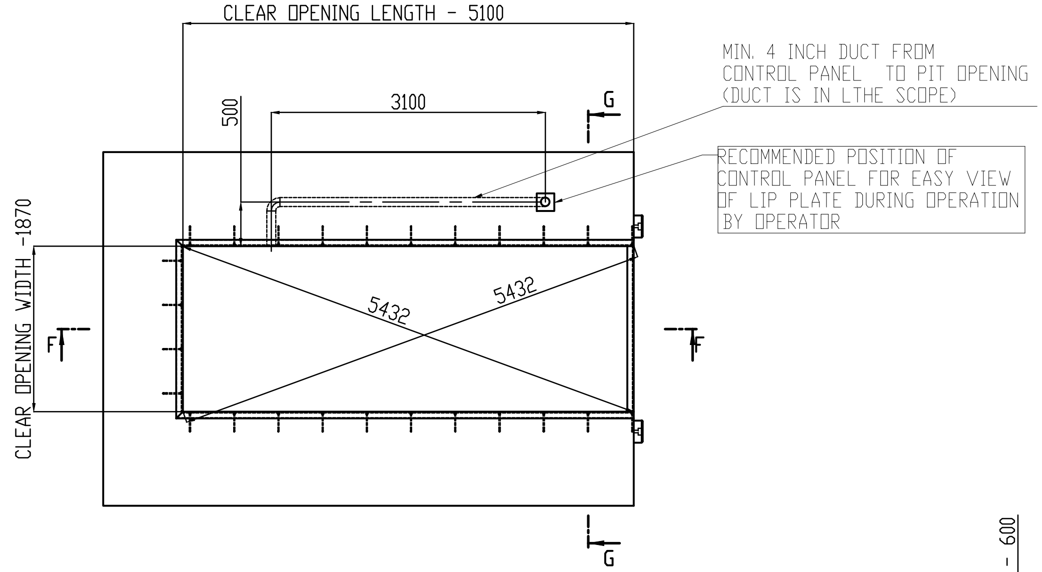
OWNER	:	**HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)**
PMC	:	**ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**
UNIT	:	**504**
PMC JOB NO.	:	**B016**
EQUIPMENT DESCRIPTION	:	**DOCK LEVELER**
EQUIPMENT TAG	:	**504-LZ-930A/B**
P. R. NO.	:	**BO16-RUF-LT-504-MP-PR-0014**
VENDOR NAME	:	**NIHVA TECHNOLOGIES PVT LTD**
VENDOR DOCUMENT NO.	:	**NIHVA_20-21_0158**

DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	**REVIEW STATUS**					
	LTHE			**EIL**		
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☑ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)			☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)		
ISSUED FOR ☐ INFORMATION ☑ REVIEW ☐ CONSTRUCTION ☐ AS BUILT						
	DISCIPLINE	SIGN	DAT	DISCIPLINE	SIGN	DATE

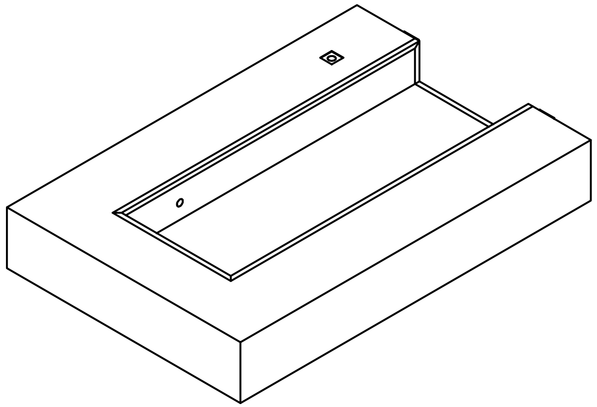
00	Issued for Information						DD-Mmm-'YY
Revision No.	**Description**	**Prepared by**	**Reviewed by**	**Approved by**	**Reviewed by**	**Approved by**	**Approved Date**

REVISION HISTORY SHEET

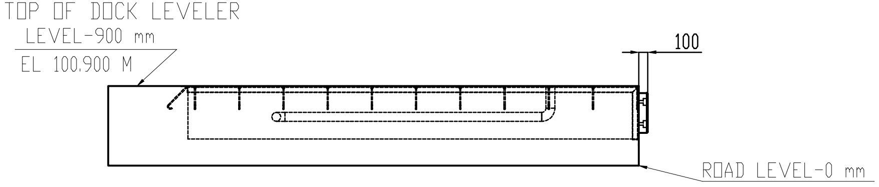
REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
01	24/09/2021	NA	ISSUED FOR APPROVAL


CLEAR OPENING LENGTH - 5100
3100
500
G
MIN. 4 INCH DUCT FROM CONTROL PANEL TO PIT OPENING (DUCT IS IN LTHE SCOPE)
RECOMMENDED POSITION OF CONTROL PANEL FOR EASY VIEW OF LIP PLATE DURING OPERATION BY OPERATOR
CLEAR OPENING WIDTH -1870
5432
5432
F
F
G

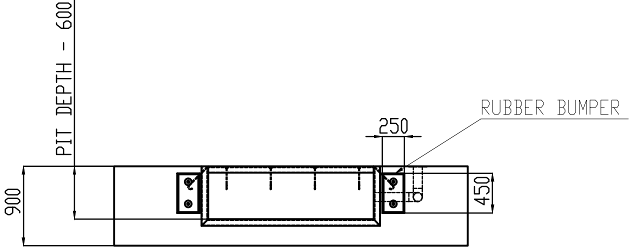
TOP VIEW OF DOCK PIT



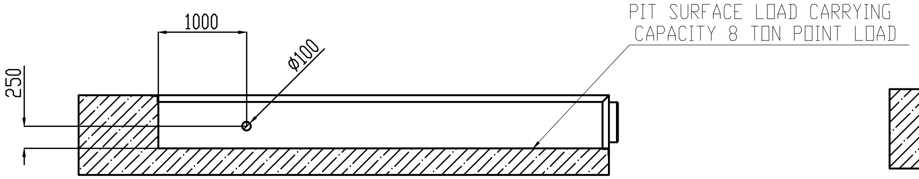
ISOMETRIC VIEW OF DOCK LEVELER PIT


TOP OF DOCK LEVELER
LEVEL-900 mm
EL 100.900 M
100
ROAD LEVEL-0 mm

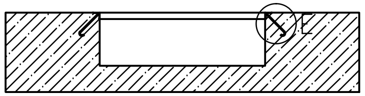
FRONT VIEW OF DOCK PIT


PIT DEPTH - 600
RUBBER BUMPER
250
900
450

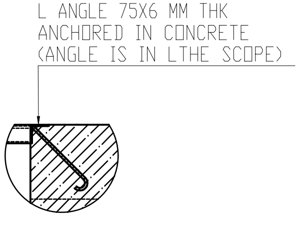
SIDE VIEW OF DOCK PIT


1000
Ø100
250
PIT SURFACE LOAD CARRYING CAPACITY 8 TON POINT LOAD

SECTION F-F
SCALE 1 : 28



SECTION G-G
SCALE 1 : 28


L ANGLE 75X6 MM THK ANCHORED IN CONCRETE (ANGLE IS IN LTHE SCOPE)

DETAIL E
SCALE 1 : 10

NOTES -

1. PIT & ALL CIVIL WORK IS IN LTHE SCOPE .
2. AT ALL EDGES OF PIT ANGLE 75X6 MM THK(ANGLE IN LTHE SCOPE) TO BE ANCHORED IN CONCRETE .
3. PIT SURFACE SHOULD BE PLAIN & LOAD CARRYING CAPACITY FOR PIT SURFACE SHOULB OF 8 TON POINT LOAD .
4. MINIMUM 4 INCH DUCT TO BE PROVIDED FROM CONTROL PANEL TO PIT OPENING BY LTHE FOR ELECTRICAL CABLE ROUTING . REQUIRED DUCT IS IN LTHE SCOPE.
5. CONTROL PANEL IS PROVIDED WITH ELECTRICAL CABLE LENGTH OF 5 METER AS STANDARD . CONTROL PANEL SHOULD BE LOCATED NEAR DOCK LEVELER FOR EASY VIEW OF LIP RESTING ON VEHICLE .

DWG. NO.

L&T - CHIYODA LIMITED

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☐ V-VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

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504-LZ-930A/B	GAD OF DOCK LEVELER	
DRAWING NO.	DRAWING TITLE	REV

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) R I
SCH. DATE OF SUBMISSION
EIL REVIEW/APPROVAL CODE:
VPTL NO.:

DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☐ AS BUILT
ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW
SIGN: DATE:

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NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: **HINDUSTAN PETROLEUM CORPORATION LIMITED**
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

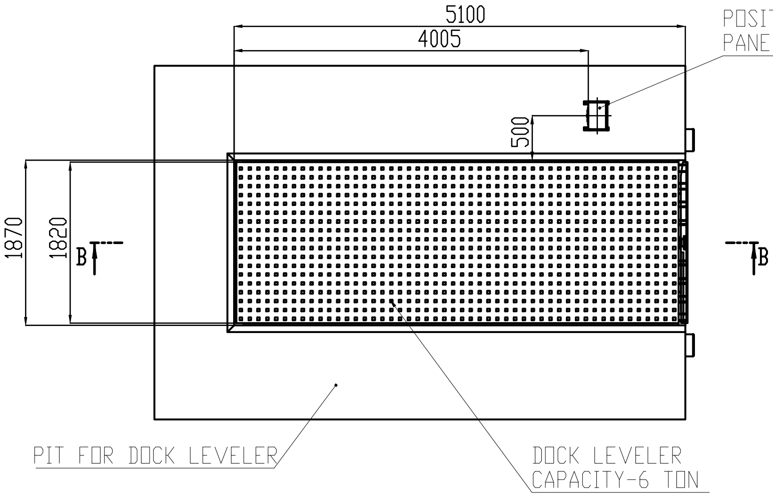
PMC : इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

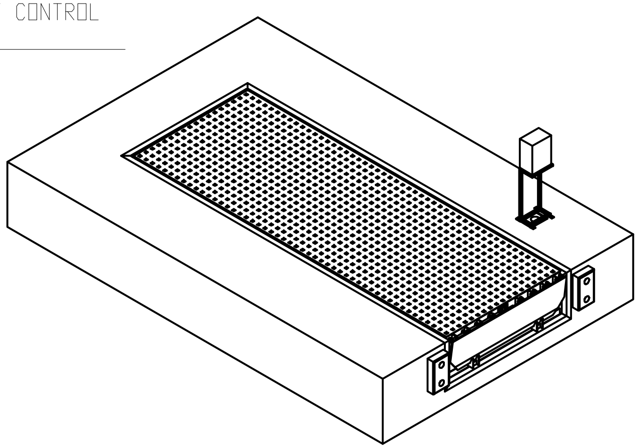
LSTK CONTRACTOR: L&T Hydrocarbon Engineering

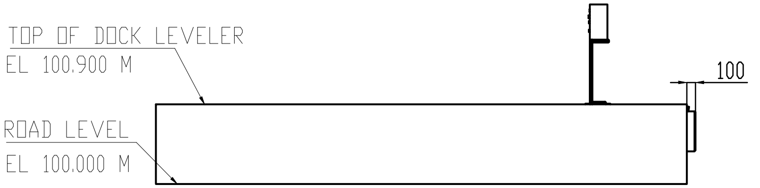
DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

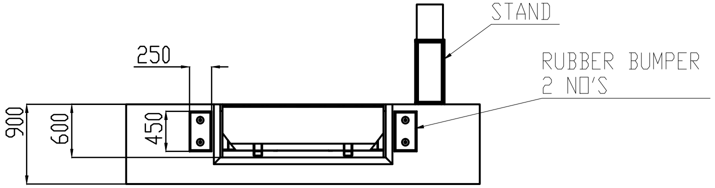
VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VMM
VENDOR DWG. NO: NIHVA_20-21_0158

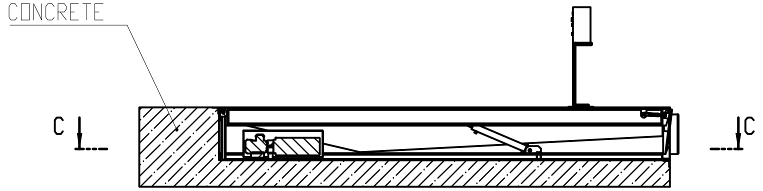
DATE 29.5.21
SCALE 1:1
TITLE: CIVIL DRAWING FOR DOCK LEVELER (504-LZ-930A/B)
DEPT: DESIGN
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: B066-RUF-LT-504-MP-DV-NTPL(1)-13005
SH. 2 OF 2
REV.00
CAD REVISE THIS DRG. USING CAD SYSTEM ONLY FILE:


5100
4005
POSITION OF CONTROL PANEL
500
1870
1820
B
B
PIT FOR DOCK LEVELER
DOCK LEVELER CAPACITY-6 TON

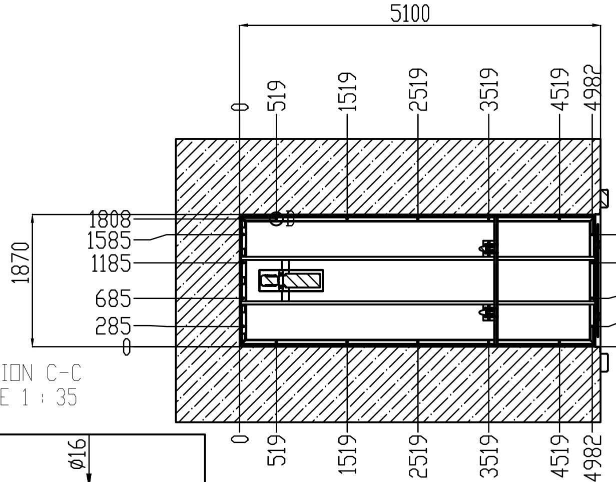



TOP OF DOCK LEVELER
EL 100.900 M
ROAD LEVEL
EL 100.000 M
100


STAND
RUBBER BUMPER
2 NO'S
250
900
600
450


CONCRETE
C
C

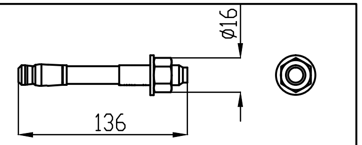
SECTION B-B
SCALE 1 : 28


5100
0
519
1519
2519
3519
4519
4982
1808
1585
1185
685
285
0
1870
1585
1185
685
285
0
D

SECTION C-C
SCALE 1 : 35

PRODUCT DESCRIPTION -

1. DOCK LEVELER - PIT MOUNTED ELECTRO-HYDRAULIC TYPE
2. CAPACITY- 6 TON POINT LOAD
3. QTY- 2 NOS
4. PRODUCT SIZE- 5100mm(L)X1820mm(W)X600mm(D)
5. PIT SIZE-5100mm(L)X1870mm(W)X600mm(D)



DETAILS OF ANCHOR WEDGE BOLT M16

DETAIL D
SCALE 1 : 35

L&T - CHIYODA LIMITED

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☐ R - DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
☐ V-VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

DRAWING NO.	DRAWING TITLE	REV
504-LZ-930A/B	GAD OF DOCK LEVELER	

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) R ☐ I ☑
SCH. DATE OF SUBMISSION

EIL REVIEW/APPROVAL CODE:
VPTL NO.:

DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☐ AS BUILT

ISSUED FOR
☐ AS BUILT ☑ INFORMATION
☐ CONSTRUCTION ☐ REVIEW

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☐ V-VOID

NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: HINDUSTAN PETROLEUM CORPORATION LIMITED
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)

PMC : ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

LSTK CONTRACTOR: L&T Hydrocarbon Engineering

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

VENDOR : NIHVA TECHNOLOGIES PVT LTD	VENDOR PO NO: REFINARY/75000-82256/VMM
	VENDOR DWG. NO: NIHVA_20-21_0158

DATE 31.05.21	SCALE 1:1	TITLE: CIVIL DRAWING FOR DOCK LEVELER (504-LZ-930A/B)	
DEPT: DESIGN			
DRN BY: SMS	PROJECT: L&T-RUF	DRG. NO.: B016-RUF-LT-504-MP-SV-NTPL(D)-13005	SH. 1 OF 2
CHK BY: VJ	PWO NO.:		REV. 00
APP BY: AK	CAD REVISE THIS DRG. USING CAD SYSTEM ONLY FILE:		